DermaFace AI — Evaluation Report

Architecture: cnn   Owner: Ali   Date: 2026-07-24

Checkpoint: dermaface_best_cnn.pt (best val macro-F1 = 0.3864, epoch 20)

Task: 4-class condition classification (acne / rosacea / redness / clear). Severity de-scoped for v1 (see docs/severity-decision.md).

Overall metrics (frozen test set)

Metric	Value
Accuracy	0.3481
Macro-F1	0.3539
Macro-Precision	0.3336
Macro-Recall	0.4437

Target vs. Actual (docs/requirements.md)

Requirement	Target	Actual	Met?
P1 Beat majority baseline	Required	0.3481 vs baseline 0.3987	False
P2 Macro-F1 (4-class)	>= 0.60	0.3539	False
P3 Per-class recall	>= 0.50 each	acne=0.06, rosacea=0.45, redness=0.42, clear=0.84	False
Fa1 Macro-F1 gap across skin-tone bands	<= 0.15	0.0636	True

Per-class results

Class	Precision	Recall	F1	Support
acne	0.2500	0.0635	0.1013	63
rosacea	0.2273	0.4545	0.3030	22
redness	0.3571	0.4167	0.3846	48
clear	0.5000	0.8400	0.6269	25

Fairness by Fitzpatrick skin-tone band

Band	Test support	Accuracy	Macro-F1
I-II	81	0.2963	0.3332

III-IV	61	0.3934	0.3674
V-VI	16	0.4375	0.3038

Confusion matrix

true \ pred	acne	rosacea	redness	clear
acne	4	15	28	16
rosacea	4	10	7	1
redness	5	19	20	4
clear	3	0	1	21

Error analysis

The Basic CNN performs at roughly chance and does not beat the majority-class baseline (accuracy 0.35 vs. 0.40), failing P1–P3. The confusion matrix explains why: the model reliably recognises clear skin (recall 0.84) but collapses on acne (recall 0.06 — only 4 of 63 acne images correct), scattering its acne predictions across redness (28), clear (16) and rosacea (15); rosacea and redness are likewise confused with each other. In effect the network has learned a "clear vs. not-clear" boundary but cannot separate the three erythema conditions. This is unsurprising — acne, rosacea and redness all present as red, inflamed skin, and a 6-layer network trained from scratch on ~1,000 images lacks the capacity and data to pick up the finer cues (papules, telangiectasia, diffuse flushing) that distinguish them. That is exactly the intended role of this model: a weak from-scratch baseline for the pretrained models (ResNet50, VGG16) to beat. If they do not improve materially on the erythema classes, the bottleneck is data, not architecture. On fairness, the macro-F1 gap across skin-tone bands is small (0.064, within the 0.15 target), but the V–VI band has only 16 test images, so that result is indicative rather than conclusive and should be revisited once more type V–VI data is available.

Notes

Auto-generated by this notebook from a real training run. Metrics computed via dermaface.training.metrics (Iva's implementation) so numbers match evaluate.py. Error-analysis section added by hand (Ali).